

ADVAIT DHARMADHIKARI

Forward Deployed Engineer · Applied AI · Agentic & Multi-Agent Systems · Enterprise LLM Deployment · RAG & Evals

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PROFESSIONAL SUMMARY

Forward Deployed Engineer who turns ambiguous customer problems into production AI systems: RAG architectures, agentic workflows and multi-agent orchestration shipped under real enterprise constraints. As Founding AI Engineer at Frensei I own end-to-end technical delivery of an AI-first compliance intelligence platform for financial-services brokers, from discovery through production deployment. That includes LLM pipelines that cut token spends 80 to 90 percent, a Claude-based signal-detection engine with full evidence-trail audit logging for regulated decisions, and a security posture that clears CASA Tier 2 and Google OAuth Restricted Scope. I came up through ML delivery inside a 300-plus aircraft airline and enterprise consulting, so I lead with "what decision are we improving?" before touching a model. Open-source author of MCP servers and evaluation tooling, and author of two published research papers on Forward Deployed Engineering and agentic AI oversight.

CORE COMPETENCIES & TECHNICAL SKILLS

AI Engineering: Forward Deployed Engineering · Applied AI · Agentic AI · Multi-agent orchestration · RAG · Semantic search · Vector databases · MCP (Model Context Protocol) · Tool and function calling · Context & Graph engineering

LLM & Evaluation: LLM pipeline design · Prompt engineering · Eval harnesses · Hallucination detection · Confidence scoring · Structured outputs · Inference cost optimization · LLMops · Claude · OpenAI · open-weight models

Production & Platform: Python · FastAPI · Docker · PostgreSQL (row-level security) · Redis · Qdrant · Neo4j · DuckDB · Supabase · OpenTelemetry observability · CI-gated test suites · REST API design · pytest / mypy / ruff

Cloud & Deployment: GCP · Azure · Multi-tenant SaaS · Hybrid and VPC deployment · OAuth 2.0 · Gmail API · Microsoft Graph

Security & Compliance: AES-256-GCM · PII masking · Prompt-injection defense · GDPR · CASA Tier 2 · Google OAuth Restricted Scope

Delivery: Technical discovery · Solution architecture · Stakeholder alignment · Executive communication · Roadmap and pricing ownership · Python · TypeScript · SQL

PROFESSIONAL EXPERIENCE

Frensei Innovation Labs (ClawbackVault AI) · Founding AI Engineer & Product Lead

Dec 2025 to Present · Dublin, Ireland

AI-first compliance intelligence platform for financial-services broker-dealers. Own full-stack AI architecture, enterprise deployment and product strategy from 0 to scale.

- Cut LLM token spend 80 to 90 percent per broker by architecting a three-stage targeted-surveillance pipeline (header scan, PII-masked body fetch, LLM signal classification) that replaced full-inbox scanning while strengthening privacy compliance.
- Enabled regulated churn decisions across 15-plus behavioral categories through a four-tier signal-detection engine on Claude Sonnet 4.6, with tiered confidence scoring and full evidence-trail audit logging.
- Cut hallucinated and inconsistent outputs by roughly 60 percent, measured on an internal eval set of labelled broker threads, by embedding evaluation into the production pipeline: eval harnesses, structured-output verification and confidence thresholds calibrated against reviewed outcomes.
- Automated complex compliance-research workflows with agentic AI systems built around planning, execution, verification and feedback loops, orchestrated using LangGraph and tool / function calling.
- Built and led a cross-functional team spanning AI engineering, cybersecurity and analytics, defining the technical standards, playbooks and delivery practices from scratch.
- Hardened the platform to CASA Tier 2 and Google OAuth Restricted Scope and shipped secure multi-tenant deployment across EU-West and Asia-Pacific: AES-256-GCM at rest, PII masking ahead of every LLM call, prompt-injection protection, GDPR data-portability export, Supabase row-level tenant isolation, and Gmail API and Microsoft Graph integration over OAuth 2.0.
- Partnered directly with the Founder/CEO to take the platform from MVP toward scale, owning the four-phase technical roadmap and the pricing-tier architecture.

Stack: Claude Sonnet 4.6 · LLM pipelines · Agentic AI · RAG · Semantic search · LangGraph · Python · Supabase · Gmail API · Microsoft Graph · OAuth 2.0 · AES-256-GCM · GDPR

IndiGo (InterGlobe Aviation Ltd) · Business Analyst & ML Engineer

Jun 2023 to Aug 2025 · Gurugram, India

ML-driven optimization and decision intelligence across a 300-plus aircraft fleet at one of the world's largest airlines. Promoted into the ML engineering role within six months; owned the bridge between data science and executive decision-making.

- Delivered 20 percent operational cost savings across a 300-plus aircraft fleet by designing and deploying aircraft zero-fuel-weight prediction and on-time-performance cost models into production.

- Improved cross-functional decision quality by roughly 40 percent through production-grade ML and BI insight frameworks built in cloud-native environments (GCP, Azure) for operations teams and leadership.
- Ran technical discovery directly with Business Technology and operations stakeholders inside the airline, converting ambiguous business problems into ML requirements, a technical roadmap and deployed systems.
- Reduced decision latency for operations leadership by re-architecting reporting logic and model parameters around the specific decisions each output drove.

Stack: Python · GCP · Azure · ML modeling · Feature engineering · Business intelligence · Stakeholder analytics

CHRIST Consulting · Business Development Analyst (Team Lead)

Jun 2021 to Jun 2022 · India

- Drove a 25 percent increase in new-business acquisition by leading a 15-member analyst team across market-entry and growth-strategy engagements, converting analysis into executive-level recommendations.
- Lifted renewals and upsells by roughly 15 percent through data-driven proposals and client-facing analytical assessments.

Teaching & Mentorship

- Teaching Assistant, Digital Technologies in Business, University College Dublin (Jan to May 2026): postgraduate tutorials on AI in business, digital transformation and data-driven decision-making, alongside full-time engineering at Frensei.
- Business Analytics & ML Mentor, topmate.io (2025 to 2026): 1:1 coaching on LLM application development and technical positioning for professionals moving into AI engineering roles.

Earlier experience: Data science and analytics internships at IndiGo, LetsGrowMore, The Sparks Foundation and EntryLevel (2020 to 2023).

RESEARCH & PUBLICATIONS

[Forward Deployed Engineering: A Systems Engineering Perspective](#) · preprint, Zenodo, 2026

First-author preprint establishing Forward Deployed Engineering as a systems-engineering discipline, analysed against ISO/IEC/IEEE 15288 lifecycle processes through a multivocal literature review, proposing a cross-organizational deployment lifecycle for enterprise AI.

Focus: Forward Deployed Engineering · Production AI systems · Enterprise AI architecture · Multivocal literature review · ISO/IEC/IEEE 15288

[What Agent Traces Cannot Tell You: Evidentiary Adequacy of Runtime Records for Agentic AI Oversight](#) · preprint, Zenodo, 2026

First-author paper and [open benchmark](#) testing whether agent runtime records can support the findings of fact AI oversight requires. Across 1,200 agent trajectories, six determination families and four record conditions, correctly resolved determinations rise from 16.3 percent for action logs and OpenTelemetry GenAI spans to 100 percent for a labelled substrate, with weaker records failing silently.

Focus: Agentic AI oversight · AI observability · OpenTelemetry GenAI · Evidentiary adequacy · Benchmark design · EU AI Act record-keeping

Production AI Systems Engineering · ongoing open research programme

Archive bridging academia and practice: context engineering, enterprise agent architecture and reliability, AI evaluation, AI observability.

AI PROJECTS & OPEN SOURCE

[EADIP: Enterprise Autonomous Decision Intelligence Platform](#) · open-source

A governed multi-agent platform that investigates business questions end-to-end (plan, retrieve, query, verify, report). Shipped across 12 delivery phases with CI-blocking accuracy, safety and reliability gates (274 unit and integration tests). An independent Verification Agent re-derives every numeric claim before it can enter a report, every side-effecting action is blocked pending human approval, tenant isolation is enforced with Postgres row-level security, and a GraphRAG metric-lineage layer grounds "why" questions in the data.

Stack: Python · FastAPI · LangGraph-pattern orchestration · Postgres (RLS) · Qdrant · Neo4j · Redis · NL-to-SQL safety gate · OpenTelemetry

[Veritas](#) · self-hosted MCP server, open-source

An MCP server that turns Claude into a rigorous, hypothesis-driven data investigator: builds and falsifies MECE hypothesis trees over local CSV, Parquet and Excel data in DuckDB, enforces deterministic "receipts-or-it-didn't-happen" verification so every numeric claim traces to an executed artifact, and runs autonomous opportunity and risk discovery with Benjamini-Hochberg false-discovery-rate suppression, scored against a public eval suite of planted-cause datasets.

Stack: Python · MCP (Model Context Protocol) · DuckDB · Claude Desktop / Claude Code · Docker · pytest / mypy / ruff

Published Python packages (PyPI):

HalluciMap, hallucination-risk scoring and segment-level triage for LLM outputs · **FraudGuard**, fraud and anomaly classification pipeline · **FinFeatures**, financial and time-series feature engineering · **Finance AutoML**, automated model selection for financial datasets · **ESGProfiler**, ESG signal structuring and scoring · **MLOpter**, ML pipeline and hyperparameter optimization.

Stack: Python · scikit-learn · pandas / NumPy · feature engineering · AutoML · LLM APIs · evaluation heuristics

EDUCATION

MSc, Business Analytics · UCD Michael Smurfit Graduate Business School · Sep 2025 to Aug 2026

PG Professional Certification, AI & Machine Learning · IIT Kanpur · Oct 2023 to Sep 2024

BBA, Business Analytics & Data Science · Christ University, Bangalore · Jun 2020 to Jun 2023

Honors: Global Leadership Programme (awarded by the Dean of the UCD College of Business) and UCD Advantage Award, UCD Smurfit, 2026.